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Secure Cloud-Based Diabetes Classification with Machine Learning

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***Abstract*— With the continuous advancement of medical devices, a substantial amount of data is generated within the healthcare environment daily. Consequently, cloud computing technology is being employed to securely store, process, and manage these vast volumes of data, safeguarding them from potential attacks. This paper focuses on diabetes classification using machine learning (ML) algorithms with a secured cloud computing environment by combining the Elliptic Curve Cryptography (ECC) and the Advanced Encryption Standard (AES) method to ensure authentication and data integrity. Additionally, we performed a comparative analysis of diabetes prediction classification using five diverse ML algorithms. The results show that the random forest (RF) classification method achieves the best results among K-Nearest Neighbors (KNN), Naïve Bayes (NB), Diction tree (DT), and support vector machine (SVM) classifiers, with an accuracy of 90.4%, as well as the evaluation of security in the cloud involves analyzing the encryption and decryption times for records of different file sizes, ranging from 20 KB to 50 KB. The results obtained demonstrate that the proposed system surpasses other state-of-the-art systems in terms of performance. Therefore, the proposed model successfully establishes an efficient remote health monitoring system that relies on secure data storage and management on cloud-based platforms.**

Keywords— machine learning, diabetes, classification, prediction, healthcare system, cloud computing.

I. INTRODUCTION

The progress in the healthcare industry and the evolution of healthcare devices result in the generation of vast volumes of data on a regular basis, and the data acquired from these devices is multidimensional and varied [1]. Diabetes is a disease that is getting more severe and morbid in every advanced and developing country [2]. Inadequate insulin production by the pancreas leads to elevated blood sugar levels, which can damage vital organs like the eyes, nerves, heart, and kidneys [3]. A global study by Fitzmaurice et al. [4] estimates that diabetes prevalence was 8.8% in 2017 and is projected to rise to 9.9% by 2045.

While early diagnosis allows for management and control, there's currently no cure for diabetes. The classification of diabetes data poses challenges because of the intricate and interdependent structure of a large volume of medical data [5]. Furthermore, the outcomes of diabetes classification are significantly affected by the existence of numerous outliers in the dataset, as well as the existence of missing values [6]. Hence, the size and dimensionality of health databases are rapidly expanding. It is challenging to manage the file system with rising data volume as healthcare databases grow. As a result, handling medical data becomes a top priority for healthcare providers [7]. As a result, Cloud computing assists in the protection, sharing, management, and storage of electronic health records (EHRs) [8]. Enabling the sharing of healthcare information can offer practical solutions for enhancing public access to information, streamlining healthcare management processes for the general population, and ensuring ongoing healthcare reporting.

Despite the numerous benefits mentioned, several obstacles must still be overcome. The act of sharing electronic health information within an organization's cloud environment typically gives rise to various privacy challenges [9]. When it comes to storing health data on the cloud, it is of utmost importance to safeguard the information in a way that prevents the cloud from acquiring any knowledge or insight into the dataset. Consequently, the protection of medical information within the cloud infrastructure is paramount. Because health information is extremely critical, even minor changes to it can lead to inaccurate medical diagnoses [10]. As a result, an encryption technique is necessary to improve data security and privacy by securing medical data without revealing critical information. To safeguard data privacy in the cloud, encryption techniques are often used. This process scrambles the data before it's uploaded, making it unreadable even for the cloud storage provider itself [11].

Security in the cloud depends on encryption techniques, where data is scrambled into a secret code and unlocked with unique keys. In the study conducted by Chen et al. [12], they suggested employing a combination of AES and ECC to enhance system security. Our research builds upon the concept of combining AES and ECC for secure data transmission, as seen in previous work. However, we propose a novel hybrid model called AES-ECC that addresses cloud storage specifically. This approach eliminates the need for third-party involvement while prioritizing two key aspects: reducing data key size for efficiency and maintaining robust security within a shorter processing timeframe.

Therefore, Healthcare service providers create a vast amount of heterogeneous data and information regularly, making conventional methods of analyzing and processing it challenging. ML approaches to aid in the effective analysis of this data for actionable insights. Predicting an individual's risk and sensitivity to chronic disease is an essential task. Diagnosing chronic diseases saves money on medical bills and lowers the chance of more serious health problems. Even in high-risk scenarios where a patient is unconscious or opaque, it is critical that correct inferences may be made from

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readily detectable medical indications to enable doctors to make better judgments for patient therapy [13]. ML approaches are known to yield effective outcomes when extracting knowledge and creating predictive models from diagnostic medical datasets obtained from diabetes patients. The extraction of knowledge from such data holds promises in predicting diabetes in individuals. While several ML approaches can be utilized for predicting diabetes mellitus, selecting the appropriate approach based on specific characteristics can be challenging. In this study, we tackle this challenge by employing four well-known ML methods, namely RF, SVM, NB, KNN, and DT. These methods are applied to adult population data to predict the occurrence of diabetes mellitus. We noticed that Disease classifications become extremely important in compiling statistics on causes of disease (morbidity) and causes of death (mortality).

Users may lose control over their data if it is kept in the cloud and is accessible outside a physical location. Another security worry is the requirement for healthcare and medical organizations to have a clear agreement with the CSP that addresses security concerns. This paper introduces notable features and contributions, which can be summed up as follows:

- This paper presents AES-ECC, a new model that strengthens healthcare data security in cloud storage by reducing encryption key size without sacrificing system security or processing efficiency.
- This work proposes a disease prediction methodology using various ML techniques, evaluated through classification results.

Experiments show our new technique surpasses existing illness prediction systems in both security and privacy.

The paper's framework is as follows: Section II reviews existing literature, followed by Section III's discussion of proposed methodology. Section IV provides an overview of the method used in the study, while Section V presents the results and discussion. Concluding with a plan to address and overcome obstacles in future research outlined in section VI.

II. LITERATURE REVIEW

Table 1 compares ML approaches in healthcare. It highlights how these approaches address challenges like pre-diabetes screening, data security, and real-time analysis, while achieving improved accuracy and reduced latency.

Table 1: Related work

Reference	Key Features	Evaluation Criteria	Outcome
[26]	Identified significant features for pre-diabetes screening based on HbA1c levels.	Cross-validation score, Cohen's kappa, AUC.	Developed a tool for early-stage pre-diabetes identification.
[27]	Monitored children based on clinical, physiological, and socioeconomic characteristics.	Relevance estimation algorithm based on classifier.	Categorized children born with IUGR by monitoring blood pressure, identifying relevant attributes.
[28]	Federated Learning prevents data upload to the cloud, ensuring privacy and security.	Not explicitly mentioned.	Improved privacy and security for healthcare data using FL and Blockchain.
[29-30]	Compared accuracy of Blockchain-based approach with centralized methods.	Latency for large datasets.	Proposed model reduced latency for large datasets.

III. PROPOSED METHODOLOGY

The proposed work is elaborated in detail, as depicted in Figure 1. We leverage the publicly available PIDD dataset (UCI ML Repository) containing health records of 768 diabetic patients. The data includes 8 features and separates patients into diabetic (labeled 1) and non-diabetic (labeled 0) groups (500/268 split). To ensure data security during cloud storage and transmission, we implemented the AES-ECC encryption method.

Upon retrieval, the secured data undergoes decryption. Data pre-processing steps are then carried out, preparing the dataset for classification. The dataset, now ready for classification, is fed as input to different classifiers, namely DT, NB, SVM, KNN, and RF. The primary objective of using these classifiers is to accurately distinguish and classify the dataset into two categories: one for individuals with diabetes and the other for individuals without diabetes.

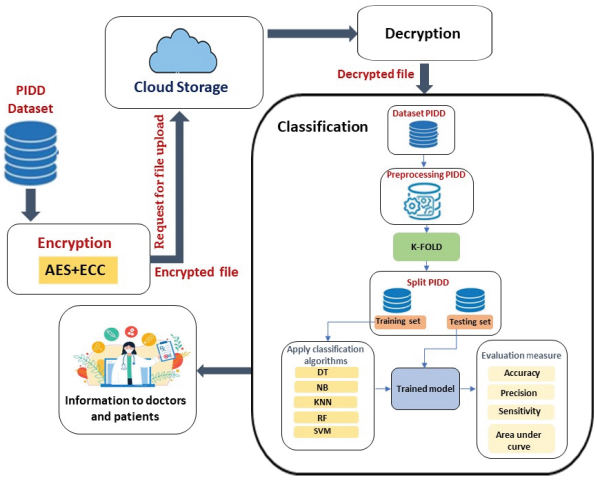


Fig. 1 The Proposed Methodology

IV. METHOD USED

A. Healthcare System

Figure 1 illustrates our AI framework for developing a ML model to predict diabetes. It details the training process for the model, allowing for updates, and its functionality in predicting diabetes incidence or prevalence. Further details on the training workflow and prediction are provided.

A (i). Dataset Pre-processing

Our algorithm incorporates data cleaning steps before processing. This includes removing outliers (data points significantly different from the rest) and filling in missing values [14-17]. Outliers can skew results, so we use the IQR (Interquartile Range) method for outlier detection (details omitted for brevity). We present the mathematical basis for outlier rejection in Equation (1).

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$$p(x) = \begin{cases} x & if\ Q1 - 1.5 * IQR \le x \le Q3 + 1.5 * IQR \\ reject & otherwise \end{cases} \quad (1)$$

Given that Q1, Q3, and the IQR are elements of $\boldsymbol{R^m}$, Q1 denotes the first quartile, Q3 denotes the third quartile, and the IQR represents the interquartile range of the utilized values.

A (ii). KCV

K-fold cross-validation (KVC) is a widely used technique for model selection and assessing classifier performance [18]. Figure 2 illustrates the visual representation of the data splitting technique utilized in this study, which follows a 5-fold cross-validation approach. The PIDD were separated into K folds, and the models were trained using the K-1 folds. The optimal hyperparameters and unreleased testing data were used (K times) To evaluate the models' performance through the outermost loop. Furthermore, to maintain a consistent % of samples in each class, KCV has been employed as the PIDD dataset consists of both positive and negative samples. Equation (2) was used to generate the final evaluation metrics [19].

$$M = \frac{1}{k} \times \sum_{n=1}^{k} P_n \pm \sqrt{\frac{\sum_{n=1}^{k} (p_n - \bar{p})^2}{k-1}} \quad (2)$$

In the above equation, M denotes the final performance metric of the classifiers, K represents the number of folds utilized in the evaluation.

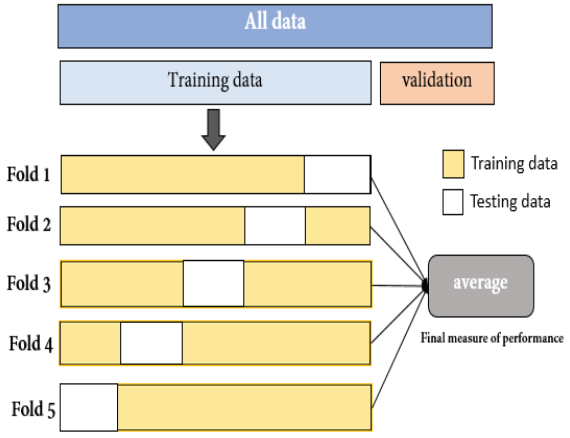


Fig. 2. 5-fold partitioned cross-validation of the implemented PID data collection

B. The Proposed Hybrid Approach (ECC and AES)

The hybrid encryption method of ECC and AES depicted in Figure 3 is highly effective for cloud storage security. This hybrid approach capitalizes on the benefits of both methods: AES offers robust encryption, while ECC keeps key sizes manageable. This synergistic effect enhances overall data security in the cloud without requiring excessively large keys [20].

This ECC-AES combination is ideal for cloud storage security and reduces the size of secure data storage [21]. Our proposed approach, leveraging a combination of ECC and AES (ECC-AES), ensures secure healthcare data sharing in the cloud. This method guarantees confidentiality, privacy, and data integrity. Additionally, it offers robust protection against unauthorized access. Even if an attacker intercept uploaded files, the strong AES encryption renders the data unreadable, safeguarding user information.

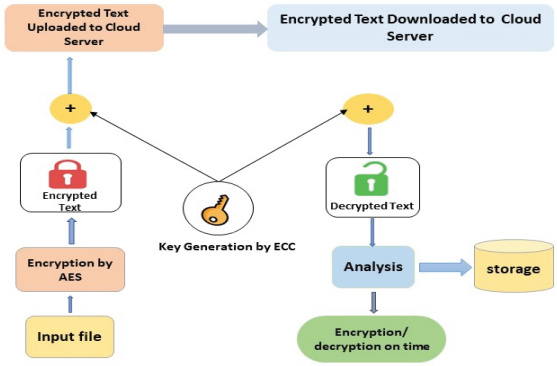


Fig. 3. Representation of AES-ECC algorithm

B (i). Algorithm for the Proposed Framework

Algorithm 1: Creating a Public Key through ECC

Input:
n: A large prime number

Output:
P: The public key

1. Choose n(i): Select a random number n(i) such that 0 < n(i) < n
2. Select Elliptic Curve Point: Choose a point C on a pre-defined elliptic curve, where C > n
3. Calculate Public Key: P = n(i) * C (point multiplication on the elliptic curve)
4. Return Public Key: Output P

Algorithm 2: AES: Encryption and Decryption

Input:
File: The file to be encrypted and decrypted

Output:
Encrypted File: The encrypted version of the original file

1. Input File Acquisition: Acquire the file to be secured (File).
2. Public Key Integration: Append the public key (generated using ECC) to the file (File + Public Key).
3. Encrypt the combined data (Combined Data) using the AES encryption algorithm with the public key (Public Key) as the encryption key. This operation generates an encrypted file. (Encrypted File = AES_Encrypt(Combined Data, Public Key))
4. Encrypted File Upload: Upload the encrypted file (Encrypted File) to the cloud server.
5. Decryption upon Download (placeholder):
   (Note: This step is included for completeness but not shown in detail due to security concerns of revealing decryption methods)
   Download the encrypted file (Encrypted File) from the server.
   Employ the public key provided by ECC to unlock the downloaded encrypted file, revealing the original data. (Decrypted File = AES_Decrypt(Encrypted File, Public Key))

B (ii). Comparisons of Various Algorithms

Depending on the site in the body of their onset, cancers are categorized into various types as shown in figure 3. In literature, n number of cancers exist that affect various body

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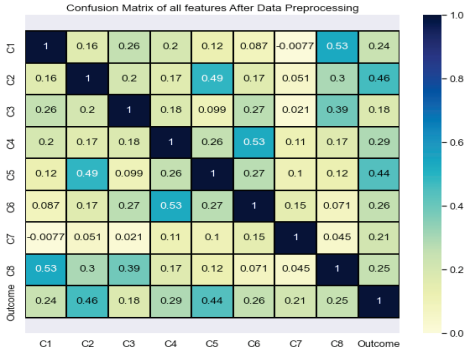
organs and lead to the worst quality of life. The five most common vital organs affected by cancers these days include breast, lungs, colorectal, prostate and stomach.

V. RESULTS AND DISCUSSION

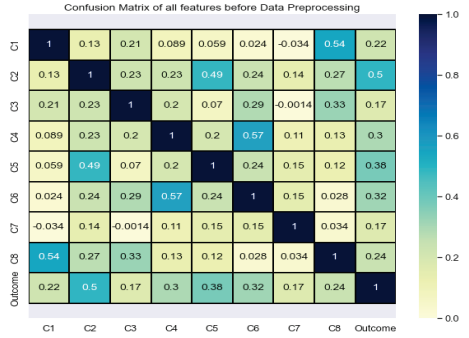
This research aims to accurately classify health data. The system effectively distinguishes healthy samples from those with disease, potentially even considering disease severity. Secure data transfer is crucial. Implemented in Python, the system utilizes PIDD datasets and evaluates its performance using metrics like accuracy, recall, precision, and AUC. Additionally, security assessments analyze encryption and decryption times, comparing them to existing methods. Experiments with various classification algorithms assess the system's ability to categorize disease instances, and the resulting data is used to evaluate the proposed technique's overall effectiveness.

A. confusion matrix

Figure 4 (replace with actual figure reference) depicts a correlation matrix that visualizes the impact of outlier removal and missing value imputation on data quality. Examining both the visual representation in Figure 4(a) and the corresponding numerical values in Figure 4(b) reveals a clear improvement in the connection between various attributes (C1, C2, etc.) and the focus outcome variable. (This paraphrase emphasizes the confirmation aspect of the analysis). This enhancement is particularly evident for attributes C3, C4, and C5, where correlation coefficients show a significant increase. This improved correlation strengthens the effectiveness of feature selection techniques that rely on correlation analysis to identify the most relevant attributes for model building.



(a)



(b)

Fig. 4. Correlation Analysis through Confusion Matrix (a) Before processing. (b) After processing.

A. Results for ML Model

To classify the instances of the disease, multiple experiments were conducted using different classification algorithms including DT, SVM, NB, KNN, and RF. A performance comparison was carried out using various ML metrics. The performance evaluation based on these metrics can be found in Table 2, verifying the performance examination. The outcomes for selecting the optimal pre-processing and ML model are presented in Table 2. This table displays the accuracy, precision, recall, and AUC values, allowing for comparison among the different approaches. The summary outlines each model's ability to achieve the highest AUC when following the proposed pipeline. It has been observed that all classifiers achieve their optimal performance when "outlier rejection (P)", "filling missing values (Q)", and "correlation-based feature selection techniques (CRB)" are applied to the PIDD. Each classifier demonstrates its highest performance when these pre-processing steps are employed. The two experiments, as shown in Table 2, show that The Accuracy of the SVM, DT, KNN, NB, and RF has improved by 6.7 %, 13.8%,17%, 1.6%, and 19.3% respectively when P, Q, and CRB. the precision improved by 11.3%, 24.7%, 26.3%, 6.6%, 29%.

Table 2. The proposed approach outperforms existing methods, with the folding technique yielding the best results.

	Classifier	Accuracy	Precision	Recall	AUC
Without pre-processing	**SVM**	0.76	0.667	0.655	0.815
	DT	0.747	0.625	0.727	0.742
	KNN	0.695	0.583	0.509	0.764
	NB	0.75	0.661	0.709	0.825
	RF	0.721	0.607	0.618	0.812
P+Q+ CRB (n-attribute=6) proposed	**SVM**	0.827	0.78	0.78	0.866
	DT	0.885	0.872	0.829	0.875
	KNN	0.853	0.846	0.805	0.918
	NB	0.766	0.727	0.585	0.859
	RF	**0.914**	**0.897**	**0.854**	**0.948**

Table 3 summarizes the top-performing models identified through our evaluation process. Figure 5 visually depicts the variation in model performance, emphasizing the Area Under the Curve (AUC) metric. Figure 5a illustrates model performance without any pre-processing applied to the data. Conversely, Figure 5b showcases the impact of appropriate pre-processing techniques on model performance.

The superiority of the RF model is evident in Figure 5b, where it surpasses DT, SVM, NB, and KNN models. This advantage is further solidified by the Area Under the Curve (AUC) values obtained by the RF classifier on the test dataset. These high AUC values indicate the RF model's strength and use in analysing this dataset.

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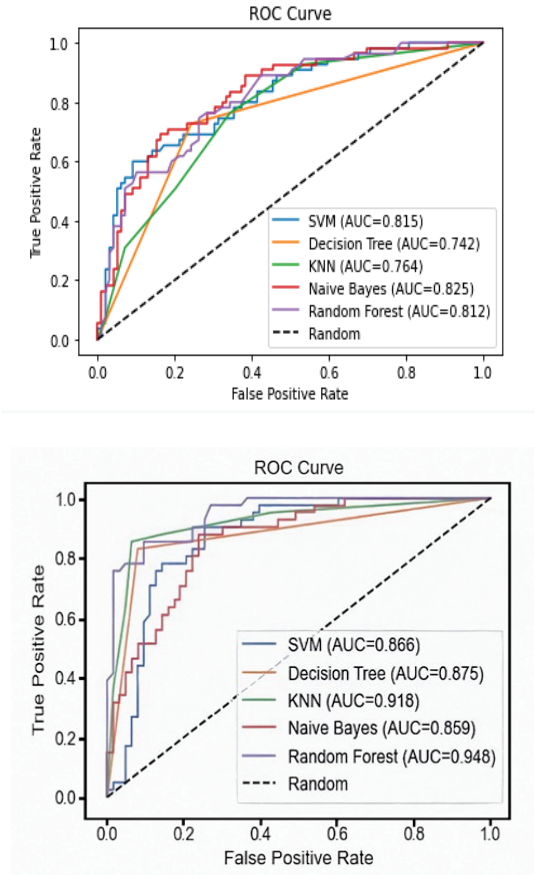


Fig. 5. (a) ROC curve of the classifiers after pre-processing, (b) ROC curve of the proposed model classifiers.

To assess the effectiveness of various ensemble learning and ML models, this research employs a comprehensive set of evaluation metrics. These metrics, including accuracy, recall, precision, and AUC, provide a detailed picture of model performance. Building on prior research in diabetes classification using the PIDD metric, this section analyzes the performance of various ML models. Our findings reveal that the RF classifier stands out, achieving superior accuracy (91.40%), precision (89.70%), recall (85.40%), and AUC (94.80%) compared to other algorithms.

Table 3 further strengthens this observation by offering a side-by-side comparison of results from prior research.

Table 3. Proposed Model Performance Comparison on PIDD Dataset

author	methodology	Accuracy (%)	Precision (%)	Recall (%)	AUC (%)
Ramesh et al. [23]	KNN	80.0	-	87.20	-
	LR	73.30	-	70.20	-
	NB	73.10	-	66.60	-
	SVM+RBF	83.20	-	87.30	-
Hanaa et al.[24]	DT	81.89	84.78	89.66	90.76
	NB	81.89	85.71	88.64	89.92
	Fuzzy KNN	90.55	91.30	95.45	90.55
	TFKNN	90.63	93.18	93.18	94.13
Aziz et al. [25]	KNN	83.12	-	74,19	-
	SVM	82.70	78.00	78.00	86.60
	DT	88.50	87.20	82.90	87.50
The proposed	KNN	85.30	84.60	80.50	92.00
	NB	77.00	73.00	59.00	86.00
	RF	**91.40**	**90.00**	**85.00**	**95.00**

B. Data Security Effectiveness Assessment

The execution of the novel AES-ECC method is assessed in comparison to existing AES, DES, and ECC algorithms in expressions of encryption/decryption time. This evaluation of performance metrics is presented and verified in Table 4 and Table 5. Additionally, a graphical representation of the performance comparison based on these metrics is depicted in Figure 6.

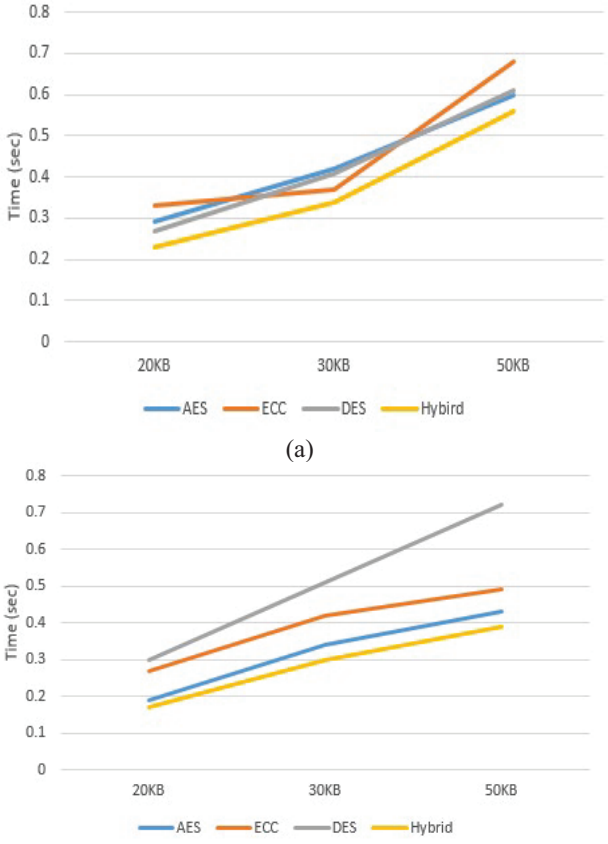


Fig. 6. Graphs of encryption (a) and decryption times (b) for the proposed technique

Evaluation confirms the proposed method's efficiency in both encryption and decryption. It outperforms existing algorithms (AES, ECC, DES) for files ranging from 20KB to 50KB, achieving faster encryption and decryption times (e.g., 0.23 seconds encryption for 20KB file compared to 0.27-0.33 seconds with other methods).

Table 4. Encryption time analysis (sec)

	20KB	30KB	50KB
AES	0.29	0.42	0.6
ECC	0.33	0.37	0.68
DES	0.27	0.41	0.61
Hybrid	**0.23**	**0.34**	**0.56**

Table 5. Decryption time analysis (sec)

	20KB	30KB	50KB
AES	0.19	0.34	0.43
ECC	0.27	0.42	0.49
DES	0.3	0.51	0.72
Hybrid	**0.17**	**0.3**	**0.39**

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VI. CONCLUSION

A critical challenge in healthcare using ML for disease prediction is balancing patient privacy with leveraging sensitive medical data. This research proposes a novel system that prioritizes data privacy yet enables prediction within existing frameworks. The system utilizes a unique combination of AES and ECC for secure data transmission. Analyses confirm the system's effectiveness in both secure transmission and disease prediction accuracy. The proposed AES-ECC approach outperforms established encryption algorithms, while the Random Forest classifier within the system achieves an impressive 91.40% accuracy, surpassing existing methods. This study presents a novel approach that excels in both disease prediction accuracy and patient privacy protection, ensuring a secure environment for healthcare data analysis.

Our research lays the groundwork for future advancements. Expanding the model's capabilities to encompass a broader spectrum of data types and exploring DL techniques hold promise for further refinement of disease prediction accuracy. However, these advancements must prioritize the implementation of robust security and privacy safeguards for patient data.

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